

Solutions to Jan 12 ex. of Math 6320

a) M is Noetherian $\Leftrightarrow M', M''$ are Noetherian

$\Rightarrow N \subset M'$ is also a submodule of M and so it is finitely gen.

If $N \subset M''$, then let $f: M \rightarrow M''$, and $\tilde{N} = f^{-1}(N)$.

Then $\tilde{N}$ is finitely generated and so $N = \tilde{N}/_{M'}$ is also finitely gen.

$\Leftarrow$ Let $N \subset M$ be a submodule, $f: N \rightarrow M''$ the induced homomorphism. Since $\ker(f) \subset M'$, one has that $\ker(f)$ is finitely generated. Since $\text{Im } f = N/\ker f \subset M''$, one has that $N/\ker f$ is finitely generated. From the short exact sequence $0 \rightarrow \ker f \rightarrow N \rightarrow N/\ker f \rightarrow 0$, one sees that N is finitely generated (eg by generators of $\ker f$ and lifts of generators of $N/\ker f$ to N).

b) Clearly $\ker u \subset \ker u^2 \subset \ker u^3 \dots$ so let

$N = \bigcup_{m \geq 0} \ker u^m$, then one has $N = \ker u^m$ for some $m \geq 0$.

So $\ker u^m = \ker u^{m+1}$ but this is impossible as

u^m is surjective and so for $0 \neq n \in \ker u$, there is an element $\tilde{n}$ with $u^m(\tilde{n}) = n$ ie $\tilde{n} \notin \ker u^m$ but $\tilde{n} \in \ker u^{m+1}$

this is a contradiction unless $\ker u = 0$. But then u is injective and surjective and so u is an isomorphism

Let $M = K[x_1, x_2, \dots, x_n, \dots]$ Define $u(x_i) = \begin{cases} 0 & \text{if } i=1 \\ x_{i-1} & \text{if } i > 1 \end{cases}$

Then u extends to a surjective map $u: M \rightarrow M$.

$\ker u \neq 0$ so u is not an isomorphism

c) $f \cdot g = 0$ $f = a_n x^n + \dots + a_0$ $g = b_m x^m + \dots + b_0$

assume that g is of smallest degree with $f \cdot g = 0$.

Then $a_n b_m = 0$ so $\deg(a_n g) < \deg g$ but $f \cdot a_n g = 0$ so $a_n g = 0$. So $a_n b_i = 0$ for $0 \leq i \leq m$. Assume $a_{n-i} g = 0$ for $0 \leq i \leq r-1$, then the coefficient of x^{m+n-r} in $f \cdot g$

$$is 0 = a_{n-r} b_m + a_{n-r+1} b_{m-1} + \dots + a_n b_{m-r} = a_{n-r} b_m$$

So $a_{n-r} g$ has smaller degree than g and $a_{n-r} g = 0$, so as above $a_{n-r} g = 0$.

We conclude that $a_i g = 0$ for $0 \leq i \leq n$. So $a_i b_m = 0$ for $0 \leq i \leq n$.

d) $x^2 + 3x + 2 = (x+1)(x+2)$ $x+2$ is irreducible in $\mathbb{Z}[X]$, $\mathbb{Z}[[X]]$
but $x+1$ is irred. in $\mathbb{Z}[X]$ and a unit in $\mathbb{Z}[[X]]$

e) $f = a_n x^n + \dots + a_0$ is a unit $\Leftrightarrow a_0$ is a unit & $a_1, \dots, a_n$ are nilpotent

$\Leftarrow a_0$ is a unit so it suffices to show that $a_0^{-1} f$ is invertible.

Write $a_0^{-1} f = 1 + a_0^{-1}(a_1 x + \dots + a_n x^n)$. Now $a_i x^i$ is nilpotent for $i \geq 1$.

As we have seen before, $n = a_0^{-1}(a_1 x + \dots + a_n x^n)$ is also nilpotent

($AC[X]$ is a commutative ring!) so $(1+n)^{-1} = 1 - n + n^2 - n^3 + n^4 - n^5 + \dots$

(a finite sum as n is nilpotent).

$\Rightarrow$ Let $g \in AC[X]$ s.t. $f g = 1$. $g = b_m x^m + \dots + b_0$ then $a_0 b_0 = 1$

so a_0 is a unit. Now $a_n b_m = 0$. We claim that $a_n^{r+1} b_{m-r} = 0$

for $0 \leq r \leq m$. By induction, assume $a_n^{i+1} b_{m-i} = 0$ $0 \leq i < r$

then $a_n^{r+1}(a_n b_{m-r} + \dots + a_{n-r} b_m) = 0$ as this is a multiple of

the coeff. of x^{m+n-r} in $f \cdot g$. So $a_n^{r+1} b_{m-r} = 0$.

But $a_n^{m+1} b_0 = 0 \Rightarrow a_n^{m+1} \cdot b_0 a_0 = a_n^{m+1} = 0 \Rightarrow a_n$ is nilpotent.

So $(f - a_n x^n) \cdot g = 1 - a_n x^n \leftarrow$ is invertible by $\Leftarrow$

so $(f - a_n x^n) \cdot h = 1$ so a_{n-1} is nilpotent $\dots$ by

induction a_i are nilpotent.